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PROJECT AND TEAM INFORMATION

Project Title

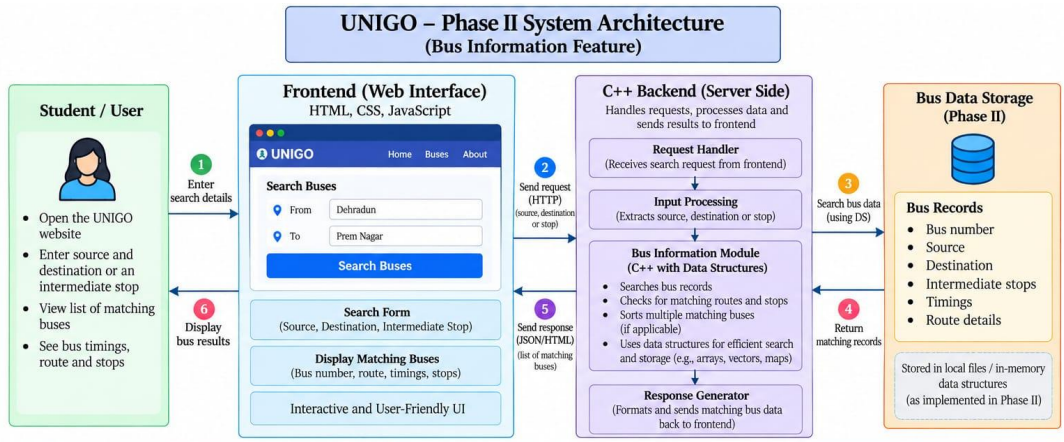
UNIGO-A Smart Campus Navigation and Student Assistance Platform

Team Name: *Team #*	*UNIGO*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

UNIGO is a website aimed at making it easier for students to find the right college bus. Students often have difficulties because they are not familiar with route numbers, the stops, or the shift times. I n Phase 2, the Bus Information feature was developed, so that students can input a destination or a bus number and then see all the buses that match their search, together with the route, the stops, the departure time, and the details regarding the campus gate. The frontend part of the system makes use of HTML, CSS, and JavaScript, while the backend is written in C++ and includes a bus-search module written in C. This module holds the bus records in structures and arrays, matches them to the students' searches, and arranges the results according to departure time. We have prepared a synthetic dataset containing the route numbers, shift times, departure times, destinations, and stops, and have tested it using sample inputs. The main goal is to give students quick and easy access to bus route information. In the future, the system will allow students to find their bus on campus when returning.



Updated Project Approach and Architecture

The system is made up of three layers: the frontend, which is composed of HTML, CSS and JavaScript, gathers the students' search queries and shows the results; the C++ backend handles the request, checks the input and passes it on to the bus module; and the C module is responsible for storing the bus records, carrying out searches and arranging the matches in order of departure time. The searching facility is used to find both the final destination and the intermediate stops which allows the users to ask by bus numbers. In some cases when many buses matches the search criteria then the result are shown in a particular order with the help of sorting especially when the departure or arrival times are same. Before displaying any result the website program handles the search cases which are empty or if the destination is unknown or the bus number is invalid or maybe the departure time is missing. For this phase we have taken bus data which has been manually checked for the accuracy. The code can be checked on the GitHub. The C module and the C++ backend needs to be linked and compiled together. Hence after all these steps the frontend needs to be connected with backe d and the testing of entire search feature from input to result will be done.

Tasks Completed (7 pts)

Task Completed	**Team Member**
The frontend development is carried out with HTML. , CSS and. JavaScriptt	Sapna Yadav
Support for bus data preparation, organization, and testing	Vanshika
The backend was developed and integrated using C++	Srishti Nahar
module development using data structures, searching, and sorting module has been noted	Anushka Pun

Challenges/Roadblocks

While developing our website UNIGO, we encountered several challenges, which include: • The key role involved linking the frontend and backend: ensuring the site could correctly execute search requests to the C++ backend and then display the appropriate results. • For this phase, the C module and the backend need to be compiled and linked correctly so that the backend can utilize the bus-searching function, implemented in C. • Destination or bus number searching : Most of the students know their destination but maybe some know their buses numbers. So in order to support both the cases we noted that the search function should go through all route stops and intermediate stops for a particular route and not just the final destination. • For a particular destination it's not compulsory that only one bus is used for that way , some multiple buses can serve the same destination. So our system should display the list of all matching buses with their numbers and destinations, along with sorting them in a order especially when two buses have similar departure or arrival times. • System and feature testing must be done: Since the C module works in a good way on its owns so it might get some issues when it is integrated with the C++ backend and our website. So in order to have a error free search process , it needs to be tested before providing.

Tasks Pending

Project Outcome/Deliverables

Unigo is a planned website helpful for students to find suitable route to their destination and campus information. Currently, we are just aiming to work on the bus route feature because students are facing huge problem about which bus to take.Through the website, users can enter their destination and then see the relevant route numbers. The platform has been developed using HTML, CSS and JavaScript, with C++ being used for the backend processing. A synthetic dataset is used to provide the route numbers, the shift times and the details of the routes. Arrays and structures are used to store the bus routes, and search algorithms have been put into place to match the routes to the user's input. The main aim of our project is to improve the accessibility of bus route information for students.

Progress Overview

The students found it difficult to identify the right bus routes, to manage their shift times, and to choose suitable destinations. As a result, the main thing we have had in mind at this stage has been the development of the bus information feature. We have produced a synthetic dataset which contains the shift times, the route numbers, the departure times, the destinations, and the bus stops. We uploaded the project to GitHub, and we are working together to improve its operational efficiency. The frontend was written using HTML, CSS and JavaScript, and we are going to implement route-search functionality by means of C++ concepts. At the moment we are looking at the coding requirements and working on the fundamental system functions. The principal difficulty has been getting the backend to work with the frontend and setting up the route-search logic in the web system. We have carried out some initial tests using sample inputs in order to examine the system's capability of identifying the correct bus routes. We are currently looking at the results to find any errors and to improve accuracy.

Codebase Information

- The project has been uploaded to GitHub, including both frontend and backend code.
- Code commits have been made according to the progress achieved so far.
- A commit has been made for uploading the synthetic dataset.
- Separate commits have also been made for frontend and backend development.
- GitHub repository: `https://github.com/Anushkapun150507/PBL_DSCPP_SEM3/tree/main`
- Problem identification and planning — Completed.
- Synthetic bus route dataset — Prepared and uploaded.
- Frontend development — HTML, CSS and JavaScript.
- Backend development —C++.
- GitHub repository and commits — Completed.
- Route searching based on user input — Completed.
- Frontend and backend integration — Completed.
- Complete testing and validation — In progress.
- Initial testing has been performed using sample inputs to check the bus route-searching functionality.
- Basic testing of the frontend, backend, and synthetic dataset is in progress to check their working and integration.

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Manual functional testing(Bus Guide)	Pass	We searched different destinations and checked that the departure point and timings shown were correct.
User interface testing (website)	Pass	We opened the website and clicked through the pages and buttons to check that they work and respond properly.
Other features (map, lost and found, reviews)	Not tested yet	These features are still in progress, so we will test them once they are ready.

Deliverables Progress

- Campus map and navigation: in progress
- Bus guide: completed, working on the website
- Lost and found: in progress
- Reviews and ratings: in progress
- Database and website: in progress,

the website frontend is ready and data set for bus navigation is ready